



Home Automation, Inc.

Hi-Fi2 by HAI Serial Control Protocol Description

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INTRODUCTION

This document describes the protocol used to communicate between an external device (personal computer, embedded controller, etc.) and the Hi-Fi2 by HAI Whole Home Audio Distribution System using a serial interface. The protocol allows the external device to monitor the status of the Hi-Fi2 system and to control its operation.

ELECTRICAL INTERFACE

The external device connects to the Hi-Fi2 system through the RS-232 serial interface built onto the Hi-Fi2 Main Assembly. The RS-232 interface is provided through a modular connector, marked "Serial Control" on the Hi-Fi2 Main Assembly.

To access the RS-232 interface, you may use the HAI Model 36A05-2 Cable Kit (this cable kit is commonly referred to as "HLC PIM to PC Modular Adapter Kit"). Plug one end of the cable into the modular connector marked "Serial Control" on the Hi-Fi2 Main Assembly. Plug the other end into the Model 36A05-2 Adapter (modular to DB-9); which in turn, connects to the DB-9 RS-232 port on the external device.

SERIAL PORT PINOUT – 6 pin modular connector

Pin 2: Transmit to system controller.

Pin 3: Receive from system controller.

Pin 5: Ground

Data is transmitted asynchronously using standard nonreturn-to-zero (NRZ) format (one start bit, eight data bits, and a single stop bit) at 9600 baud. RTS/CTS or software flow control (XON/XOFF) is not required.

All serial communications are initiated by the external serial system controller in a Command - Response format. The Hi-Fi2 will asynchronously transmit status responses to the external device whenever there is a parameter change in the system (source volume, bass, etc.). This allows the serial communications to be kept to a minimum since the external device does not need to poll the Hi-Fi2 for system state.

The particular serial commands and their use are described later in this document.

MESSAGE FORMAT AND TIMING

- 1) The data consists of printable characters with exception of the terminating characters: carriage return <CR> (or 0x0D hex) and line feed <LF> (or 0x0A hex). Upper or lower case characters may be used.
- 2) All numerical fields are coded as ASCII digit characters.
- 3) All strings are UTF-8 encoded and are enclosed in double quotes. When sending a string to the Hi-Fi2, double quotes (") and asterisks (*) must be escaped by a backslash. Backslash is represented by \.
- 4) When commands require multiple parameters, the parameters are delimited by commas.
- 5) All arguments must be specified.
- 6) Each Command string is STARTED with an ASCII "*" character and terminated by a <CR>.
- 7) Each Response string issued by the Hi-Fi2 will START with an ASCII "#" character and be terminated with a <CR><LF>.
- 8) If a command has an error in it, or does not adhere to exact command syntax, or is not recognized for any reason, the Hi-Fi2 will be echoed with a "#?RECEIVED_COMMAND<CR><LF>" string.
- 9) Whenever sending multiple commands to the Hi-Fi2, the host program should pause for 50 milliseconds between commands to prevent buffer overruns.
- 10) Whenever the status of the system has changed, a serial status message is sent indicating the new state of the controller. It does not matter what caused the state change (VSCs, serial commands, etc.) These are **unsolicited** messages and the external device needs to handle the reception of the messages to keep itself up-to-date with the Main Processor Unit. The responses to commands with "?" will also be sent unsolicited.

COMMAND EXAMPLE

This example will request the version information from the Hi-Fi2.

CMD	*VER<CR>
RSP	#VER"HAIHIFI2 FWv1.01 HWv0"<CR><LF>

This will be the format for all the commands listed in this document. The <CR> and <LF> will not be shown with the commands in this document but MUST be assumed.

COMMANDS AND RESPONSES

REQUEST SYSTEM VERSION INFORMATION

QUERY CMD	*VER
RSP	#VER"HAIHIFI2 FWvx.yy HWvz"
Command Parameters	
x	The major firmware version. Always 1 digit.
yy	The minor firmware version. Always 2 digits.
z	The hardware version. Always 1 digit.

CONFIGURE SOURCE NAMES

SET CMD	*SsNAME"new name"
QUERY CMD	*SsNAME?
Command Parameters	
s	The source number from 1-8. 0 for all sources.
"new name"	The new name for this source. Up to 16 characters in length.
RSP	#SsNAME"new name"

CONFIGURE ZONE NAMES

SET CMD	*ZzNAME"new name"
QUERY CMD	*ZzNAME?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
"new name"	The new name for this zone. Up to 16 characters in length.
RSP	#ZzNAME"new name"

CONFIGURE GROUP NAMES

SET CMD	*GgNAME"new name"
QUERY CMD	*GgNAME?
Command Parameters	
g	The group number from 1-8. 0 for all groups.
"new name"	The new name for this group. Up to 16 characters in length.
RSP	#GgNAME"new name"

CONFIGURE TRANSLATED LANGUAGE STRINGS

To put the system into a different language, translations for all of the built-in strings must be sent to the system.

SET CMD	*LSxSTR"translated string"
QUERY CMD	*LSxSTR?
Command Parameters	
x	The string number from the table below
"translated string"	The translated string to be used in place of the string in the table below. Up to 40 characters.
RSP	#LSxSTR"translated string"
Table of Language Specific Strings	
0	Balance
1	Bass
2	Treble
3	Brightness
4	Select Zone
5	Source Enable
6	Source Gain
7	IR Passthrough
8	Initial Volume
9	Page Volume
10	Enabled
11	Disabled
12	Loudness
13	IR
14-100	Undefined

CONFIGURE ZONE ENABLE/DISABLE

Zones can be enabled and disabled. Disabled zones cannot be controlled from the VSC or through the external control interfaces. All zones are enabled by default.

SET CMD	*ZzENABLEx
QUERY CMD	*ZzENABLE?
Command Parameters	
s	The zone number from 1-16
x	0 – disable the zone, 1 enable the zone
RSP	#ZzENABLEx

CONFIGURE ZONE TREBLE

SET CMD	*ZzTREBx
QUERY CMD	*ZzTREB?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
x	The treble setting from –10 to 10.
RSP	#ZzTREBx

CONFIGURE ZONE TREBLE UP/DOWN

SET CMD	*ZzTREB[+/-]
QUERY CMD	*ZzTREB?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
+/-	+ to increase treble, - to decrease treble
RSP	#ZzTREBx

CONFIGURE ZONE BASS

SET CMD	*ZzBASSx
QUERY CMD	*ZzBASS?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
x	The bass setting from -10 to 10
RSP	#ZzBASSx

CONFIGURE ZONE BASS UP/DOWN

SET CMD	*ZzBASS[+-]
QUERY CMD	*ZzBASS?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
+/-	+ to increase bass, - to decrease bass
RSP	#ZzBASSx

CONFIGURE ZONE BALANCE

SET CMD	*ZzBALx
QUERY CMD	*ZzBAL?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
x	The balance setting from -10 (left) to 10 (right)
RSP	#ZzBALx

CONFIGURE ZONE BALANCE LEFT/RIGHT

SET CMD	*ZzBAL[+-]
QUERY CMD	*ZzBAL?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
+/-	+ to shift balance right, - to shift balance left
RSP	#ZzBALx

CONFIGURE LOUDNESS

SET CMD	*ZzLOUDNESSx
QUERY CMD	*ZzLOUDNESS?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
x	0 – disable loudness compensation, 1 enable loudness compensation.
RSP	#ZzLOUDNESSx

CONFIGURE SOURCE ENABLE/DISABLE

Sources can be enabled and disabled. Disabled sources cannot be selected from the VSC or through the external control interfaces. At least 1 source must be enabled in the system.

SET CMD	*SsENABLEx
QUERY CMD	*SsENABLE?
Command Parameters	
s	The source number from 1-8
x	0 – disable the source, 1 enable the source
RSP	#SsENABLEx

CONFIGURE SOURCE GAIN

SET CMD	*SsINGAINx
QUERY CMD	*SsINGAIN?
Command Parameters	
s	The source number from 1-8
x	The input gain setting from 0 to 10
RSP	#SsINGAINx

CONFIGURE ZONE INITIAL VOLUME

This command can be used to set the “turn-on” volume for a zone.

SET CMD	*ZzINIVOLx
QUERY CMD	*ZzINIVOL?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
x	The volume level from 1 to 100.
RSP	#ZzINIVOLx

CONFIGURE ZONE PAGE VOLUME

This command can be used to set the volume for a zone during a system page event. When a page is active, all zones in the system will switch to source 8 at the specified volume; regardless of whether source 8 is enabled.

SET CMD	*ZzPGVOLx
QUERY CMD	*ZzPGVOL?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
x	The volume level from 1 to 100
RSP	#ZzPGVOLx

CONFIGURE ZONE GROUP

This command can be used to put a zone into a group. The behavior for the group is controlled by a different command.

SET CMD	*ZzGROUPx
QUERY CMD	*ZzGROUP?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
x	The group number from 1-8 (0 for no group association)
RSP	#ZzGROUPx

CONFIGURE GROUP BEHAVIOR

This command can be used to define the behavior of the group. Zones are put into groups with a different command.

SET CMD	*GgTYPEx
QUERY CMD	*GgTYPE?
Command Parameters	
g	The group number from 1-8
x	The behavior of the group. 0 – all zones in group stay on same source 1 – all zones in group stay on same source, same power state, and same relative volume (init volume can be different). 2 – all zones in group stay on same source, same power state, same relative volume (init volume can be different), and same EQ.
RSP	#GgTYPEx

RESTART THE PROCESSOR

This command will restart the processor in the Hi-Fi2t. The VER response will be broadcast when the unit finishes restarting.

QUERY CMD	*RESTART
RSP	#RESTART

CONFIGURE NETWORK – DHCP SETTING

After changing between DHCP and static IP address, the system must be restarted using the *RESTART command before the settings will take effect.

SET CMD	*NETDHCPx
QUERY CMD	*NETDHCP?
Command Parameters	
x	0 – DHCP disabled, 1 DHCP enabled
RSP	#NETDHCPx

CONFIGURE NETWORK – STATIC IP

After configuring the network for a static IP address, the system must be restarted using the *RESTART command before the settings will take effect.

SET CMD	*NETIP"aaa.bbb.ccc.ddd","eee.fff.ggg.hhh","iii.jjj.kkk.iii"
QUERY CMD	*NETIP?
Command Parameters	
"aaa.bbb.ccc.ddd"	IP address for the device to use, this only works if DHCP is disabled.
"eee.fff.ggg.hhh"	Subnet mask for the device to use, this only works if DHCP is disabled.
"iii.jjj.kkk.iii"	Default gateway for the device to use, this only works if DHCP is disabled.
RSP	#NETIP"aaa.bbb.ccc.ddd","eee.fff.ggg.hhh","iii.jjj.kkk.iii"

CONFIGURE NETWORK – DEVICE NAME

SET CMD	*NETNAME"new name"
QUERY CMD	*NETNAME?
Command Parameters	
"new name"	The new name for the device. Up to 16 characters.
RSP	#NETNAME"new name"

CONFIGURE NETWORK – TCP PORT

After changing the TCP port, the system must be restarted using the *RESTART command before the settings will take effect.

SET CMD	*NETPORTx
QUERY CMD	*NETPORT?
Command Parameters	
x	The new port number to be used for incoming TCP connections (0-65535). Default is 23.
RSP	#NETPORTx

CONFIGURE NETWORK – CONFIGURATION QUERY

SET CMD	This is a query command only.
QUERY CMD	*NETCONFIG?
RSP	#NETMAC"ab:cd:ef:gh:ij:kl " #NETNAME"new name" #NETDHCPx #NETIP"aaa.bbb.ccc.ddd", "eee.fff.ggg.hhh", "iii.jjj.kkk.ddd" #NETPORTx

CONFIGURATION RESET

This command can be used to reset the device configuration to factory defaults. The *RESTART command must be issued before all changes will take effect.

CMD	*CFGRESET
RSP	#CFGRESET

ZONE POWER ON/OFF

SET CMD	*ZzPOWERx
QUERY CMD	*ZzPOWER?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
x	The power state, 0 for off, 1 for on, 2 for toggle
RSP	#ZzPOWERx

ZONE VOLUME SET

SET CMD	*ZzVOLUMEx
QUERY CMD	*ZzVOLUME?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
x	The volume level from 1 to 100
RSP	#ZzVOLUMEx

ZONE VOLUME UP/DOWN

SET CMD	*ZzVOLUME[+~]x
QUERY CMD	*ZzVOLUME?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
x	The number of steps to increase/decrease the volume.
RSP	#ZzVOLUMEx

ZONE MUTE ON/OFF

SET CMD	*ZzMUTE~
QUERY CMD	*ZzMUTE?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
x	The mute state, 0 for off, 1 for on, 2 for toggle
RSP	#ZzMUTE~

ZONE SOURCE SET

SET CMD	*ZzSRCx
QUERY CMD	*ZzSRC?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
x	The source number from 1-8
RSP	#ZzSRCx

ZONE SOURCE PREVIOUS/NEXT

SET CMD	*ZzSRC[+~]
QUERY CMD	*ZzSRC?
Command Parameters	
z	The zone number from 1-16. 0 for all zones.
+/-	+ for next source, - for previous source
RSP	#ZzSRCx

ALL ZONES OFF NOTIFICATION

This command will be issued 5 minutes after the last zone in the system is turned off. It indicates that the system is in a power-saving mode.

RSP	#ALLOFF
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SOURCE METADATA SET

SET CMD	*SsDISPLINE"new metadata"
QUERY CMD	*SsDISPLINE?
Command Parameters	
s	The source number from 1-8.
"new metadata"	The new metadata for this source. Up to 80 characters.
RSP	#SsDISPLINE"new metadata"

SOURCE BUTTON COMMAND

This command can be used for a zone to send a transport control to the active source. The VSC can send play, pause, play/pause, previous, and next. If a controller sends this command to the Hi-Fi2, the Hi-Fi2 will forward it to all listeners.

SET CMD	*ZzBUTTONx
Command Parameters	
z	The zone number of the controller that is initiating this command from 1-16.
x	The button id: x=1: play x=2: pause x=3: play/pause x=4: previous track x=5: next track
RSP	#ZzSsBUTTONx

SYSTEM PAGE ON/OFF

This command can be used to initiate a system page event. When a page is active, all zones in the system will switch to source 8, regardless of whether source 8 is enabled. The external (hardware) page cannot be terminated by software.

SET CMD	*PAGE_x
QUERY CMD	*PAGE?
Command Parameters	
x	The page state. 0 for off, 1 for on, 2 for toggle
RSP	#PAGE_x
x	The page state. 0 for off, 1 for on, 3 for externally (hardware) initiated